

Sewon Kim

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OBJECTIVE

Optimal Control and Reinforcement Learning. My research goal is to build a broad-sense control framework for robotic systems, inspired and enhanced by model-based and learning-based methods.

EDUCATION

University of Seoul

B.E. in Electrical and Computer Engineering, Minor in Mathematics

Advisor: [Gyunghoon Park](#), [Dohyun Kwon](#)

Mar 2022 – Aug 2027 (Expected)

Seoul, South Korea

EXPERIENCE

AI Lead Engineer at InBody AI Scale Team, InBody

InBody AI Scale [A4]

Jan 2024 – Jan 2025

Seoul, South Korea

Ftenth Lead Researcher at Control and Dynamic System Lab, University of Seoul

Ftenth [A3], Advisor: [Gyunghoon Park](#)

Mar 2023 – Dec 2024

Seoul, South Korea

PUBLICATIONS

C.1 **Sewon Kim**, Hyunwoo Lee, Jeongyeol Kwon, and Dohyun Kwon. *First-Order Methods for Trilevel Optimization*. Under review at NeurIPS 2026.

PROJECTS

A.1 Autonomous air purifying robot agent

Aug 2025 – Sep 2026

4th Place (/200) (\$2,200) — SK NAMUHX A1: Autonomous Navigation Path Optimization

Independently implemented full-stack autonomous navigation for indoor air purifying robots purely in NumPy, without using open-source code, covering MCL, MPPI, and TSP.

A.2 Ftenth

Mar 2024 – Dec 2024

5th Place (/12) — 22nd Ftenth Autonomous Grand Prix at CDC Milan 2024

Led the project with 2 undergrads and implemented a full-stack physical autonomous navigation system for racing completely from scratch, covering SLAM, MCL, Opti, MPPI.

A.3 Cart pole system design and control by PID and NMPC

Jul 2024 – Sep 2024

3rd Place (/25) (\$220) — UOS ECE Innovation Fair

Independently developed a physical Cart-Pole system for \$272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.

A.4 InBody AI Scale

Jan 2024 – Jan 2025

CES Las Vegas 2025

Developed height estimation model (accuracy 99.4%) and face recognition system of "InBody AI Scale", fully automated body composition analyzer that reduces measurement time from 2 min to 40 sec.

A.5 ECG data-driven cardiovascular diagnosis device

Jul 2023 – Jul 2023

3rd Place (/12) (\$1450) — InBody Hackathon 2023

Developed a fast prototype.

A.6 Autonomous mission mobile robot

Jul 2023 – Jul 2023

3rd Place (/20) (\$220) — Object Detection-Based Solar Tracking Robot Challenge, Korea University

A.7 Sidaeting

Mar 2022 – Jun 2022

1,700 User Base

Implemented a matching algorithm that combinatorially optimizes collective happiness from users' ideal type data and developed the frontend for "Sidaeting", a dating service launched by UOSLIFE.

SKILLS

- **Languages:** Korean, English
- **Technical Languages:** C, C++, Java, JavaScript, Julia, Python, SQL, HTML, CSS
- **Technical Frameworks:** Isaac Sim·Lab, ROS1·2 (Melodic, Noetic, Foxy, Humble, Jazzy), MuJoCo, PyTorch